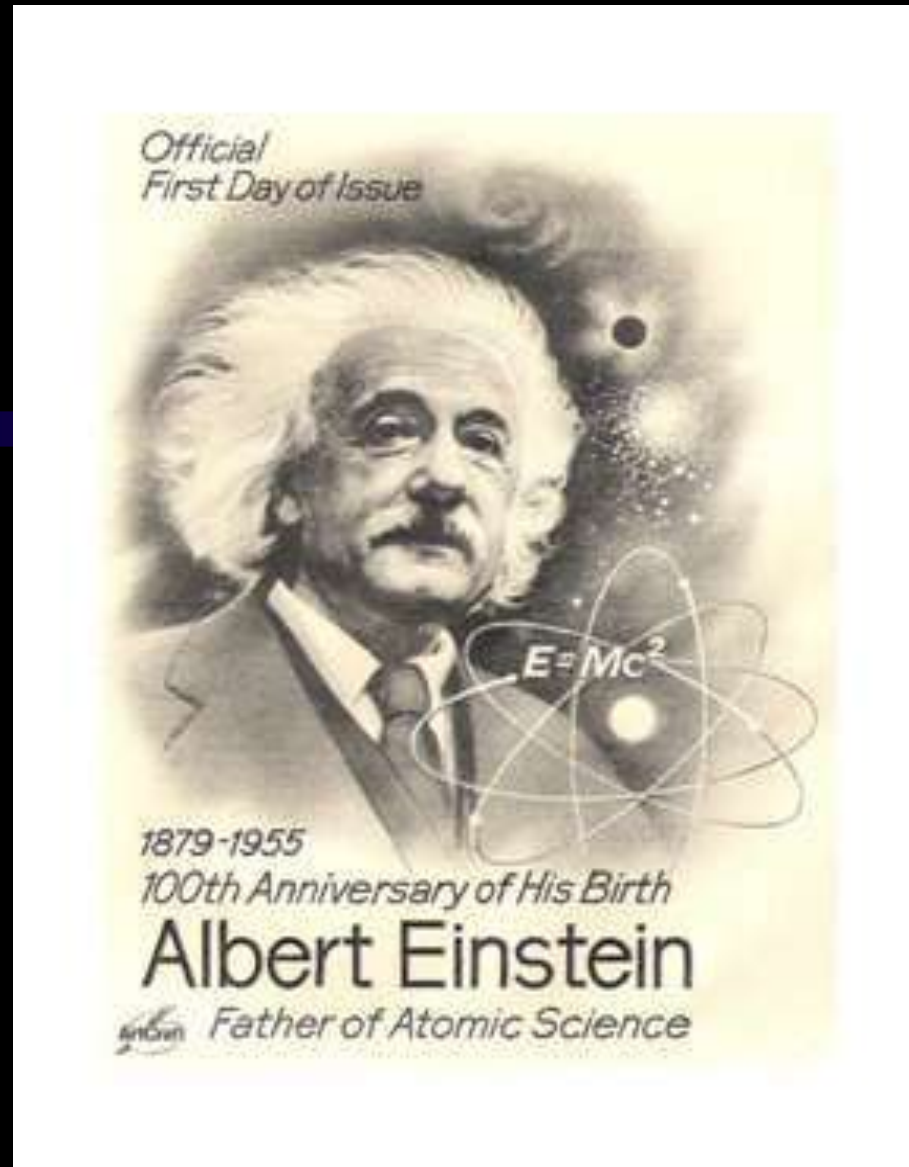
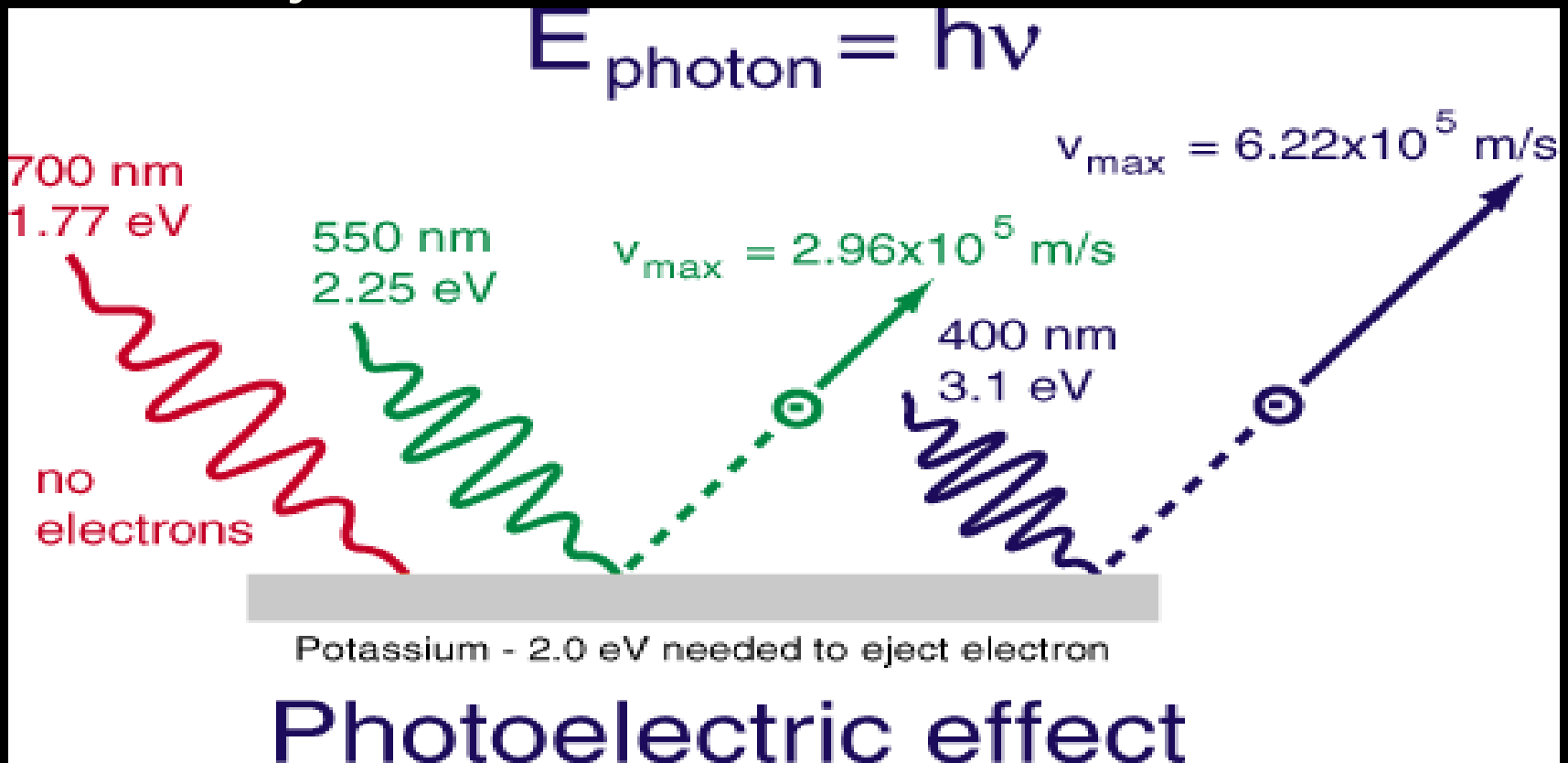


Photoelectric effect



INTRODUCTION

- First observed by **Heinrich Hertz in 1887**, the phenomenon is also known as the Hertz effect.
- Hertz observed and then showed that electrodes illuminated with ultraviolet light create electric sparks more easily.

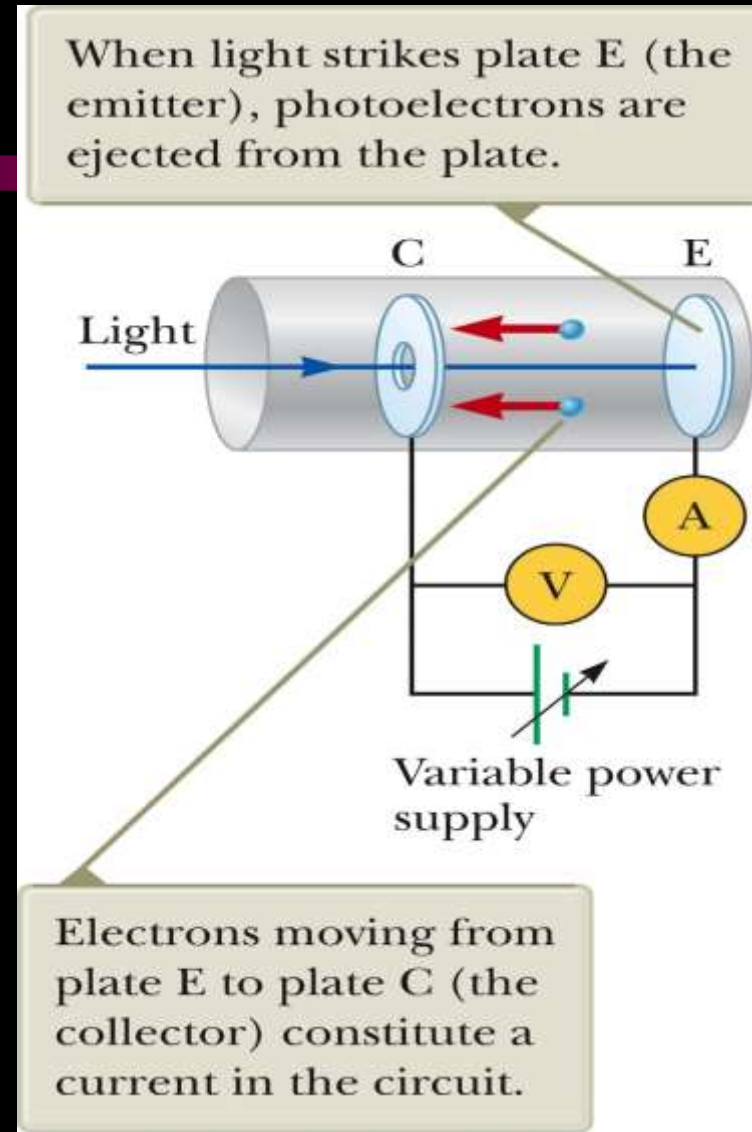


Photoelectric Effect

- The **photoelectric effect** occurs when light incident on certain metallic surfaces causes electrons to be emitted from those surfaces.
 - The emitted electrons are called **photoelectrons**.
 - They are no different than other electrons.
 - The name is given because of their ejection from a metal by light in the photoelectric effect

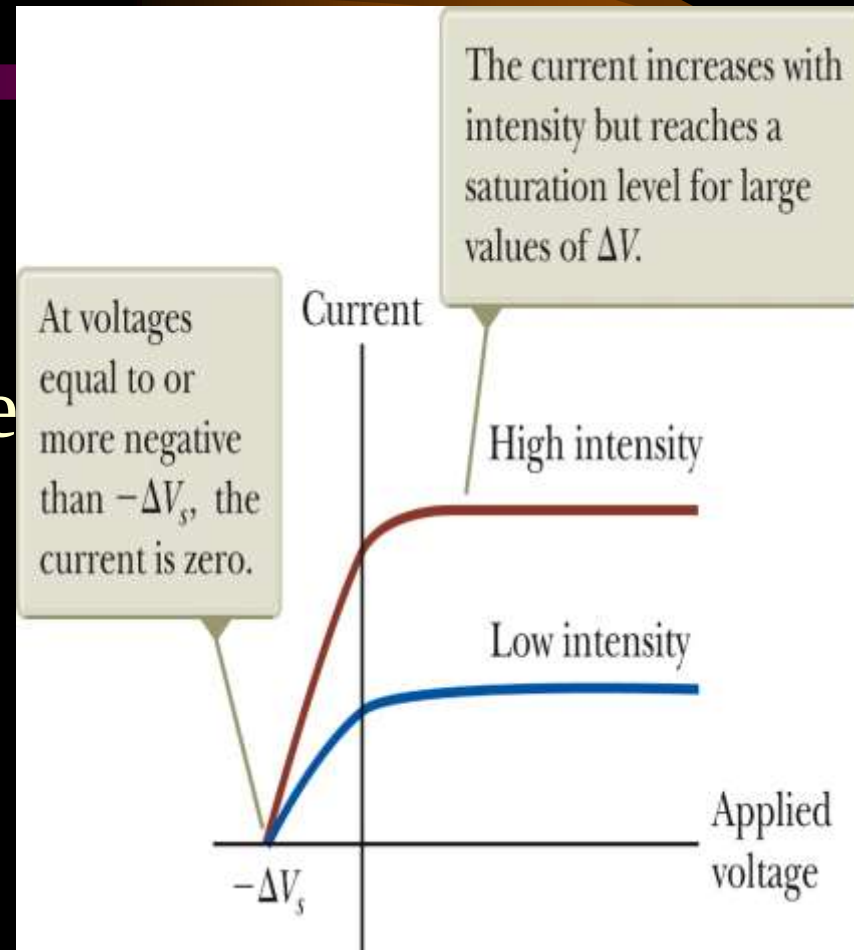
Photoelectric Effect Apparatus

- When the tube is kept in the dark, the ammeter reads zero.
- When plate E is illuminated by light having an appropriate wavelength, a current is detected by the ammeter.
- The current arises from photoelectrons emitted from the negative plate and collected at the positive plate.



Photoelectric Effect, Results

- At large values of ΔV , the current reaches a maximum value.
 - All the electrons emitted at E are collected at C .
- The maximum current increases as the intensity of the incident light increases.
- When ΔV is negative, the current drops.
- When ΔV is equal to or more negative than ΔV_s , the current is zero.



Photoelectric Effect Feature 1

- Dependence of photoelectron kinetic energy on light intensity

- *Classical Prediction*

- Electrons should absorb energy continually from the electromagnetic waves.
 - As the light intensity incident on the metal is increased, the electrons should be ejected with more kinetic energy.

- *Experimental Result*

- The maximum kinetic energy is independent of light intensity.
 - The maximum kinetic energy is proportional to the stopping potential (ΔV_s).

Photoelectric Effect Feature 2

- Time interval between incidence of light and ejection of photoelectrons

- *Classical Prediction*

- At low light intensities, a measurable time interval should pass between the instant the light is turned on and the time an electron is ejected from the metal.
- This time interval is required for the electron to absorb the incident radiation before it acquires enough energy to escape from the metal.

- *Experimental Result*

- Electrons are emitted almost instantaneously, even at very low light intensities.

Photoelectric Effect Feature 3

- Dependence of ejection of electrons on light frequency

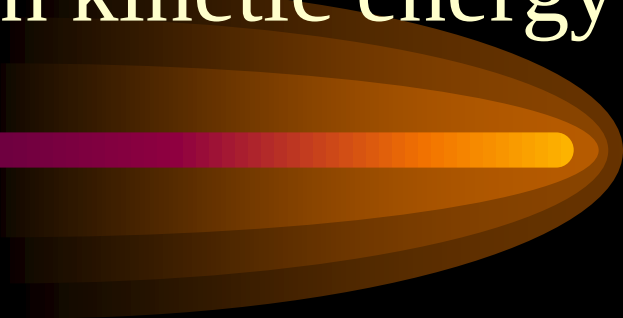
- *Classical Prediction*

- Electrons should be ejected at any frequency as long as the light intensity is high enough.

- *Experimental Result*

- No electrons are emitted if the incident light falls below some **cutoff frequency**, f_c .
 - The cutoff frequency is characteristic of the material being illuminated.
 - No electrons are ejected below the cutoff frequency regardless of intensity.

Photoelectric Effect Feature 4

- Dependence of photoelectron kinetic energy on light frequency
 - *Classical Prediction*
 - There should be no relationship between the frequency of the light and the electric kinetic energy.
 - The kinetic energy should be related to the intensity of the light.
 - *Experimental Result*
 - The maximum kinetic energy of the photoelectrons increases with increasing light frequency.
- 

Photoelectric Effect Features, Summary

- The experimental results contradict all four classical predictions.
- Einstein extended Planck's concept of quantization to electromagnetic waves.
- All electromagnetic radiation of frequency f from any source can be considered a stream of quanta, now called *photons*.
- Each photon has an energy E and moves at the speed of light in a vacuum.

$$E = hf$$

- A photon of incident light gives all its energy to a single electron in the metal.

Photoelectric Effect, Work Function

- Electrons ejected from the surface of the metal and not making collisions with other metal atoms before escaping possess the maximum kinetic energy $K_{\max}$.
- $K_{\max} = hf - \phi$
 - ϕ is called the work function of the metal.
 - The work function represents the minimum energy with which an electron is bound in the metal.

Some Work Function Values

TABLE 40.1

*Work Functions
of Selected Metals*

Metal	ϕ (eV)
Na	2.46
Al	4.08
Fe	4.50
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14

Note: Values are typical for metals listed. Actual values may vary depending on whether the metal is a single crystal or polycrystalline. Values may also depend on the face from which electrons are ejected from crystalline metals. Furthermore, different experimental procedures may produce differing values.

Photon Model Explanation of the Photoelectric Effect

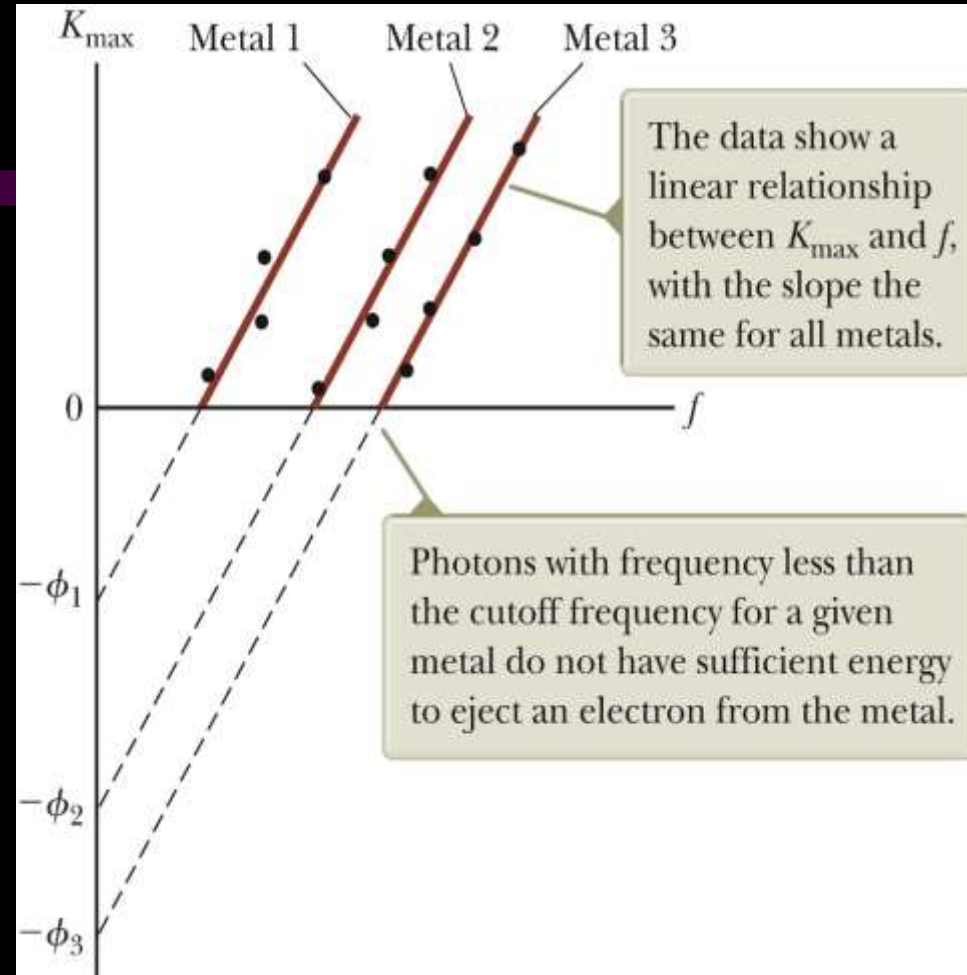
- Dependence of photoelectron kinetic energy on light intensity
 - $K_{\max}$ is independent of light intensity.
 - K depends on the light frequency and the work function.
- Time interval between incidence of light and ejection of the photoelectron
 - Each photon can have enough energy to eject an electron immediately.
- Dependence of ejection of electrons on light frequency
 - There is a failure to observe photoelectric effect below a certain cutoff frequency, which indicates the photon must have more energy than the work function in order to eject an electron.
 - Without enough energy, an electron cannot be ejected, regardless of the fact that many photons per unit time are incident on the metal in a very intense light beam.

Photon Model Explanation of the Photoelectric Effect, cont.

- Dependence of photoelectron kinetic energy on light frequency
 - Since $K_{max} = hf - \phi$
 - A photon of higher frequency carries more energy.
 - A photoelectron is ejected with higher kinetic energy.
 - Once the energy of the work function is exceeded
 - There is a linear relationship between the maximum electron kinetic energy and the frequency.

Cutoff Frequency

- The lines show the linear relationship between K and f .
- The slope of each line is h .
- The x -intercept is the **cutoff frequency**.
 - This is the frequency below which no photoelectrons are emitted.



Cutoff Frequency and Wavelength

- The cutoff frequency is related to the work function through $f_c = \phi / h$.
- The cutoff frequency corresponds to a **cutoff wavelength**.

$$\lambda_c = \frac{c}{f_c} = \frac{hc}{\phi}$$

- Wavelengths greater than λ_c incident on a material having a work function ϕ do not result in the emission of photoelectrons.

The photo electric effect

- This suitable frequency is called Threshold Frequency and the corresponding wavelength is called threshold wavelength.
- • Work function: The work function is the energy required to remove an electron from the highest filled level in the Fermi distribution of a solid.
- $$W = h\nu_0$$

Photoelectric effect

- Photoelectric effect is directly proportional to intensity.
- If the frequency of the incident light is less than the threshold frequency then no electron ejected, no matter what the intensity.
- The maximum kinetic energy of the electrons depend on the frequency of the incident light.
- The electrons were emitted immediately - no time lag.

